to this situation, went as follows. We stationary observers can easily measure the circumference and the radius of the spinning ride. For instance, to measure the circumference we can carefully lay out a ruler-head to tail-alongside the ride's spinning girth; for its radius we can similarly use the head-to-tail method working our way from the central axle of the ride to its outer rim. As we anticipate from high-school geometry, we find that their ratio is two times the number pi—about 6.28—just as it is for any circle drawn on a flat sheet of paper. But what do things look like from the perspective of someone on the ride itself?

To find out, we ask Slim and Jim, who are currently enjoying a spin on the Tornado, to take a few measurements for us. We toss one of our rulers to Slim, who sets out to measure the circumference of the ride, and another to Jim, who sets out to measure the radius. To get the clearest perspective, let's take a bird's-eye view of the ride, as in Figure 3.1. We have adorned this snapshot of the ride with an arrow that indicates the momentary direction of motion at each point. As Slim begins to measure the circumference, we immediately see from our bird's-eye perspective that he is going to get a different answer than we did. As he lays the ruler out along the circumference, we notice that the ruler's length is shortened. This is nothing but the Lorentz contraction discussed in Chapter 2, in which the length of an object appears shortened along the direction of its motion. A shorter ruler means that he will have to lay it out—head to tail—more times to traverse the whole circumference. Since he still considers the ruler to be one foot long (since there is no relative motion between Slim and his ruler, he perceives it as having its usual length of one foot), this means that Slim will measure a longer circumference than did we.

What about the radius? Well, Jim also uses the head-to-tail method to find the length of a radial strut, and from our bird's-eye view we see that he is going to find the same answer as we did. The reason is that the ruler is not pointing along the instantaneous direction of the motion of the ride (as it is when measuring the circumference). Instead, it is pointed at a ninety-degree angle to the motion, and therefore it is not contracted along its length. Jim will therefore find exactly the same radial length as we did.